

FinnCenter — Sơ đồ cơ sở dữ liệu

PostgreSQL • Prisma schema • 44 bảng • A4 ngang (297 × 210 mm)

Tài khoản & bảo mậtKhóa học & học tậpKiểm tra & AITuyển dụng giảng viênThanh toán & doanh thuThông báo & email

PHYSICAL DATA MODEL

PK: khóa chính • FK: khóa ngoại • UQ: duy nhất • ?: nullable

Course LMS PK id: String name: String description: String thumbnail: String category: String? level: String? duration: String? lessons: Int status: CourseStatus deleteRequestedFromStatus: CourseStatus? createdAt: DateTime updatedAt: DateTime price: Float languageId: String? → LearningLanguage.id instructorId: String? → User.id	TeacherApplication HR PK id: String FK userId: String → User.id FK languageId: String → LearningLanguage.id FK examLocationId: String? FK examLocationAddress: String? FK examLocationNote: String? FK recruitmentRoundId: String? → RecruitmentRound.id FK status: TeacherAppStatus FK attemptNo: Int FK entranceTestId: String? → Test.id FK entranceAttemptId: String? FK answerState: Json? FK startedAt: DateTime FK submittedAt: DateTime FK reviewedAt: DateTime FK reviewedById: String? → User.id FK rejectionReason: String? FK violationCount: Int FK failureReason: String? FK antiCheatAcknowledgedAt: DateTime? FK proctorSessionId: String? FK proctorHeartbeatAt: DateTime? FK proctorHeartbeatSequence: Int FK currentQuestionInstanceId: String? FK sequentialCompletedAt: DateTime? FK entrancePassingScore: Float? FK serverTimestamp: DateTime FK metadata: Json? FK incidentId: String? UQ (applicationId,incidentId)	TeacherEntranceQuestionInstance HR PK id: String FK applicationId: String → TeacherApplication.id FK sourceQuestionId: String FK sequence: Int FK int type: QuestionType FK content: String FK audioUrl: String? FK hint: String? FK score: Float FK preparationTimeSeconds: Int? FK answerTimeSeconds: Int? FK answerOptions: Json? FK scoringData: Json? FK status: QuestionInstanceStatus FK revealedAt: DateTime FK answerStartsAt: DateTime? FK deadlineAt: DateTime? FK answerState: String? FK answerRevision: Int FK finalAnswer: String? FK finalizedAt: DateTime? FK transitionTokenHash: String? FK transitionTokenExpiresAt: DateTime? FK speakingAudioUrl: String? FK speakingAudioHash: String? FK speakingAudioMime: String? FK speakingAudioBytes: Int? FK speakingMediaStatus: SpeakingMediaStatus FK speakingTranscript: String? FK speakingProcessingTokenHash: String? FK speakingProcessingTokenExpiresAt: DateTime? FK speakingProcessingError: String? FK speakingProcessingAttempts: Int FK contentHash: String FK createdAt: DateTime FK updatedAt: DateTime UQ (applicationId,sequence)	TeacherRevenueWithdrawal PAY PK id: String FK teacherId: String → User.id FK amount: Int FK bankName: String FK accountNumber: String FK accountName: String FK bankBranch: String? FK bankVerificationStatus: BankVerificationStatus FK status: WithdrawalStatus FK note: String? FK transferTransactionCode: String? FK processedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	AIAssessment TEST PK id: String FK userId: String → User.id FK courselId: String? → Course.id FK type: String FK taskType: String? FK title: String? FK prompt: String? FK submissionText: String? FK audioUrl: String? FK durationSeconds: Int? FK score: Float FK maxScore: Float FK bandSystem: String FK bandLevel: String FK bandScore: Float FK criteria: Json FK feedback: Json FK mistakes: Json? FK improvements: Json? FK sampleAnswer: String? FK startedAt: DateTime FK submittedAt: DateTime FK createdAt: DateTime FK updatedAt: DateTime	Payment PAY PK id: String FK-UQ orderId: String? → Order.id FK userId: String → User.id FK courselId: String? → Course.id FK purpose: String? FK pointAmount: Int? FK provider: String FK bnRef: String FK amount: Int FK status: String FK orderInfo: String? FK bankCode: String? FK transactionNo: String? FK responseCode: String? FK transactionStatus: String? FK payDate: String? FK rawResponse: Json? FK expiresAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	Test TEST PK id: String FK courselId: String? → Course.id FK languageId: String? → LearningLanguage.id FK moduleId: String? → Module.id FK lessonId: String? → Lesson.id FK kind: TestKind FK assessmentMode: TestAssessmentMode FK name: String FK description: String? FK maxScore: Float FK passingScore: Float FK maxAttempts: Int? FK timeLimit: Int? FK shuffleQuestions: Boolean FK materialTitle: String? FK materialContent: String? FK materialUrl: String? FK materialType: String? FK materialData: Json? FK createdAt: DateTime FK updatedAt: DateTime	
AntiCheatLog HR PK id: String FK applicationId: String → TeacherApplication.id FK testAttemptId: String? → TestAttempt.id FK eventType: String FK detail: String? FK severity: Int FK questionSeconds: Int? FK confidence: Float? FK clientTimestamp: DateTime? FK serverTimestamp: DateTime FK metadata: Json? FK incidentId: String? UQ (applicationId,incidentId)	CourseReport PAY PK id: String FK courselId: String → Course.id FK reporterId: String → User.id FK lessonId: String? → Lesson.id FK category: CourseReportCategory FK title: String FK description: String FK truthfulConfirmed: Boolean FK status: CourseReportStatus FK response: String? FK respondedById: String? → User.id FK attempts: Int FK resendAvailableAt: DateTime FK requestUp: String? FK deviceFingerprint: String? FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccountChangeLog PAY PK id: String FK teacherId: String → User.id FK previousBankName: String? FK previousAccountNumberMasked: String? FK previousAccountName: String? FK previousBranch: String? FK nextBankName: String FK nextAccountNumberMasked: String FK nextAccountName: String FK nextBranch: String? FK requestUp: String? FK deviceFingerprint: String? FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccountChangeOtp PAY PK id: String FK teacherId: String → User.id FK email: String FK codeHash: String FK payload: Json FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	CourseRefundRequest PAY PK id: String FK studentId: String → User.id FK courselId: String → Course.id FK-UQ orderIdItemId: String → OrderItem.id FK amount: Int FK reason: String FK status: RefundStatus FK adminNote: String? FK reviewedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime FK refundMethod: String	UserInvitation AUTH PK id: String FK email: String FK username: String FK role: Role FK languageId: String? → LearningLanguage.id FK tokenHash: String FK invitedById: String → User.id FK-UQ AcceptedUserId: String? → User.id FK expiresAt: DateTime FK acceptedAt: DateTime? FK revokedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	Question TEST PK id: String FK testId: String → Test.id FK type: QuestionType FK content: String FK audioUrl: String? FK order: Int FK score: Float FK explanation: String? FK hint: String? FK preparationTimeSeconds: Int? FK answerTimeSeconds: Int? FK createdAt: DateTime FK updatedAt: DateTime	
TeacherBankAccountChangeLog PAY PK id: String FK teacherId: String → User.id FK previousBankName: String? FK previousAccountNumberMasked: String? FK previousAccountName: String? FK previousBranch: String? FK nextBankName: String FK nextAccountNumberMasked: String FK nextAccountName: String FK nextBranch: String? FK requestUp: String? FK deviceFingerprint: String? FK createdAt: DateTime FK updatedAt: DateTime	CourseReport PAY PK id: String FK courselId: String → Course.id FK reporterId: String → User.id FK lessonId: String? → Lesson.id FK category: CourseReportCategory FK title: String FK description: String FK truthfulConfirmed: Boolean FK status: CourseReportStatus FK response: String? FK respondedById: String? → User.id FK attempts: Int FK resendAvailableAt: DateTime FK requestUp: String? FK deviceFingerprint: String? FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccountChangeOtp PAY PK id: String FK teacherId: String → User.id FK email: String FK codeHash: String FK payload: Json FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	TeacherExamResult HR PK id: String FK-UQ applicationId: String → TeacherApplication.id FK checkedIn: Boolean FK completed: Boolean FK writingScore: Float? FK speakingScore: Float? FK listeningScore: Float? FK readingScore: Float? FK failed: Boolean FK submittedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	CourseRefundRequest PAY PK id: String FK studentId: String → User.id FK courselId: String → Course.id FK-UQ orderIdItemId: String → OrderItem.id FK amount: Int FK reason: String FK status: RefundStatus FK adminNote: String? FK reviewedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime FK refundMethod: String	User AUTH PK id: String FK username: String FK email: String FK password: String FK role: Role FK phoneNumber: String? FK isBanned: Boolean FK accountStatus: UserAccountStatus FK emailVerifiedAt: DateTime? FK authVersion: Int FK learningLanguageId: String? → LearningLanguage.id FK createdAt: DateTime FK updatedAt: DateTime	VideoWatchProgress LMS PK id: String FK userId: String → User.id FK lessonId: String → Lesson.id FK lastPositionSeconds: Float? FK durationSeconds: Float? FK lastHeartbeatAt: DateTime? FK startedAt: DateTime FK completedAt: DateTime? FK seekViolation: Boolean FK createdAt: DateTime FK updatedAt: DateTime UQ (userId,lessonId)	
TeacherBankAccountChangeOtp PAY PK id: String FK teacherId: String → User.id FK email: String FK codeHash: String FK payload: Json FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	TeacherExamResult HR PK id: String FK-UQ applicationId: String → TeacherApplication.id FK checkedIn: Boolean FK completed: Boolean FK writingScore: Float? FK speakingScore: Float? FK listeningScore: Float? FK readingScore: Float? FK failed: Boolean FK submittedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccountChangeOtp PAY PK id: String FK teacherId: String → User.id FK email: String FK codeHash: String FK payload: Json FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	TeacherCertificate HR PK id: String FK applicationId: String → TeacherApplication.id FK fileName: String FK fileUrl: String FK fileType: String FK fileSize: Int FK expiryDate: DateTime FK submittedAt: DateTime FK createdAt: DateTime FK updatedAt: DateTime	CourseRefundRequest PAY PK id: String FK studentId: String → User.id FK courselId: String → Course.id FK-UQ orderIdItemId: String → OrderItem.id FK amount: Int FK reason: String FK status: RefundStatus FK adminNote: String? FK reviewedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime FK refundMethod: String	User AUTH PK id: String FK username: String FK email: String FK password: String FK role: Role FK phoneNumber: String? FK isBanned: Boolean FK accountStatus: UserAccountStatus FK emailVerifiedAt: DateTime? FK authVersion: Int FK learningLanguageId: String? → LearningLanguage.id FK createdAt: DateTime FK updatedAt: DateTime	TrustedDevice AUTH PK id: String FK userId: String → User.id FK tokenHash: String FK label: String? FK userAgent: String? FK lastIp: String? FK verifiedAt: DateTime FK lastUsedAt: DateTime FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccount PAY PK id: String FK-UQ teacherId: String → User.id FK bankName: String FK accountNumber: String FK accountName: String FK branch: String? FK verificationStatus: BankVerificationStatus FK createdAt: DateTime FK updatedAt: DateTime
PasswordResetOtp AUTH PK id: String FK userId: String → User.id FK email: String FK codeHash: String FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	TeacherCertificate HR PK id: String FK applicationId: String → TeacherApplication.id FK fileName: String FK fileUrl: String FK fileType: String FK fileSize: Int FK expiryDate: DateTime FK submittedAt: DateTime FK createdAt: DateTime FK updatedAt: DateTime	TeacherBankAccountChangeOtp PAY PK id: String FK teacherId: String → User.id FK email: String FK codeHash: String FK payload: Json FK expiresAt: DateTime FK consumedAt: DateTime? FK attempts: Int FK createdAt: DateTime FK updatedAt: DateTime	OrderItem PAY PK id: String FK orderId: String → Order.id FK courselId: String → Course.id FK price: Float FK adminRevenue: Int FK teacherRevenue: Int FK revenueSplit: String FK createdAt: DateTime FK updatedAt: DateTime UQ (orderId,courselId)	CourseRefundRequest PAY PK id: String FK studentId: String → User.id FK courselId: String → Course.id FK-UQ orderIdItemId: String → OrderItem.id FK amount: Int FK reason: String FK status: RefundStatus FK adminNote: String? FK reviewedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime FK refundMethod: String	User AUTH PK id: String FK username: String FK email: String FK password: String FK role: Role FK phoneNumber: String? FK isBanned: Boolean FK accountStatus: UserAccountStatus FK emailVerifiedAt: DateTime? FK authVersion: Int FK learningLanguageId: String? → LearningLanguage.id FK createdAt: DateTime FK updatedAt: DateTime	Session AUTH PK id: String FK userId: String → User.id FK trustedDeviceId: String → TrustedDevice.id FK tokenHash: String FK expiresAt: DateTime FK lastSeenAt: DateTime FK createdAt: DateTime FK updatedAt: DateTime	
Answer TEST PK id: String FK questionId: String → Question.id FK content: String FK isCorrect: Boolean FK order: Int FK feedback: String?	LearningLanguage AUTH PK id: String FK userId: String FK code: String FK isActive: Boolean FK createdAt: DateTime FK updatedAt: DateTime	Order PAY PK id: String FK userId: String → User.id FK createdAt: DateTime FK updatedAt: DateTime	SystemSetting TEST PK key: String FK value: Json FK updatedAt: DateTime	CourseRefundRequest PAY PK id: String FK studentId: String → User.id FK courselId: String → Course.id FK-UQ orderIdItemId: String → OrderItem.id FK amount: Int FK reason: String FK status: RefundStatus FK adminNote: String? FK reviewedById: String? → User.id FK processedAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime FK refundMethod: String	User AUTH PK id: String FK username: String FK email: String FK password: String FK role: Role FK phoneNumber: String? FK isBanned: Boolean FK accountStatus: UserAccountStatus FK emailVerifiedAt: DateTime? FK authVersion: Int FK learningLanguageId: String? → LearningLanguage.id FK createdAt: DateTime FK updatedAt: DateTime	Notification MSG PK id: String FK userId: String → User.id FK title: String FK body: String FK readAt: DateTime? FK createdAt: DateTime FK updatedAt: DateTime	TeacherApplicationLog HR PK id: String FK applicationId: String → TeacherApplication.id FK status: TeacherAppStatus FK message: String FK actorId: String? FK createdAt: DateTime FK updatedAt: DateTime
Feedback LMS PK id: String FK userId: String → User.id FK courselId: String → Course.id							